

The Controllability of the Car-like System

자동차는 방방곡곡 누빌 수 있을까?

안 희 성

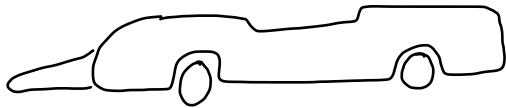
성균관대 수학과

2026. 09. 14 MIMIC

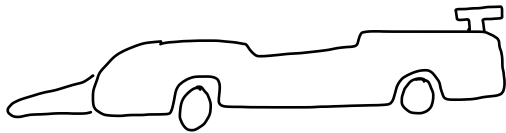
자동차 준비하는 중



자동차 준비하는 중



자동차 준비하는 중



자동차 준비하는 중



자동차 준비하는 중



자동차 준비하는 중



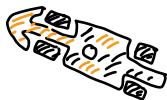
자동차 준비하는 중



자동차 준비하는 중



자동차 준비하는 중



자동차 준비하는 중



자동차 준비하는 중



자동차 준비하는 중



자동차가 특정 위치에서 특정 방향을 바라보도록

무조건 움직일 수 있을까?

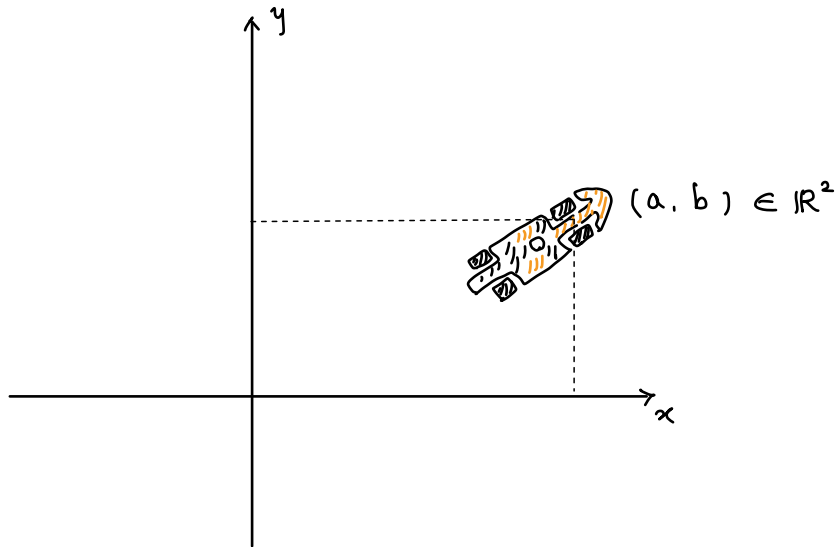
목 차

1. 모델링과 시각화
2. Rashevskii - Chow theorem
3. Orbit과 Lie Bracket

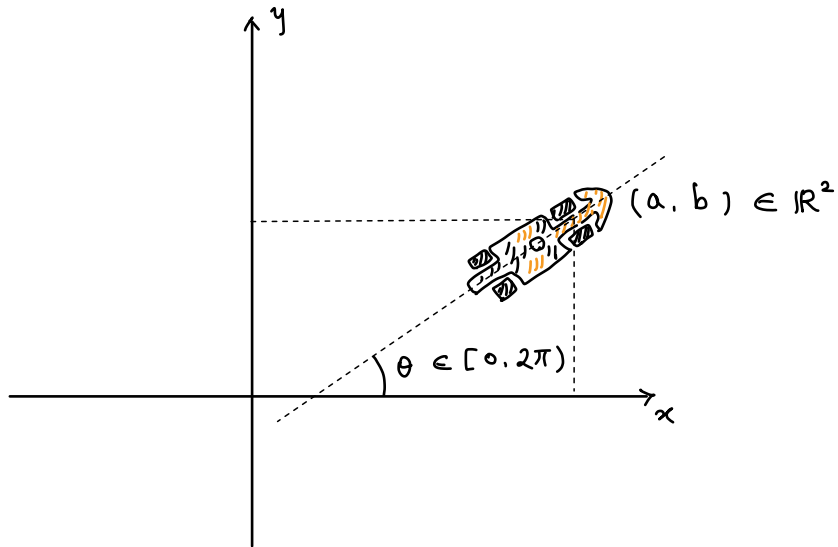
모델링과 시각화



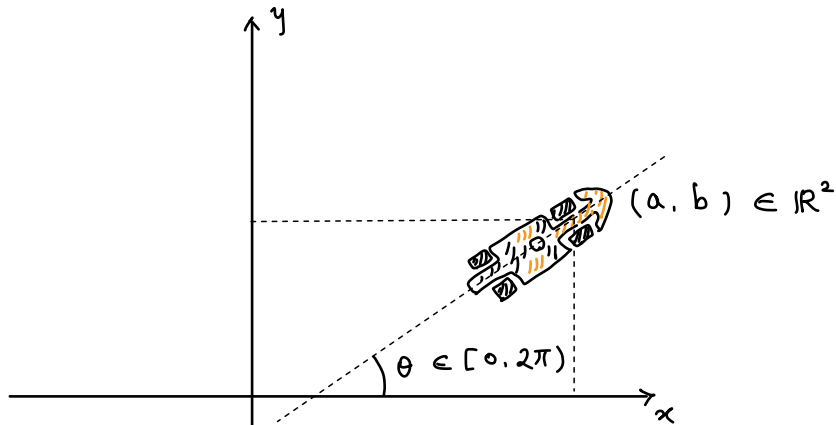
모델링과 시각화



모델링과 시각화

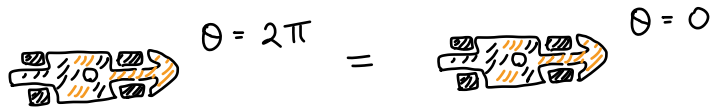


모델링과 시각화



The state is completely determined by
an element in $\mathbb{R}^2 \times [0, 2\pi)$.

모델링과 시각화

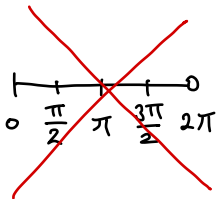


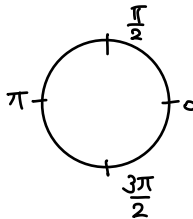
$[0, 2\pi)$ is not just a subspace of $\mathbb{R}$.

모델링과 시각화

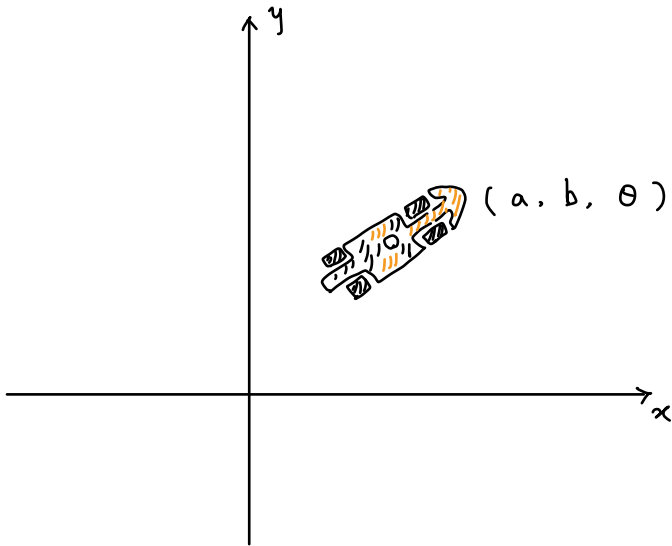
 $\theta = 2\pi$ $=$  $\theta = 0$

$[0, 2\pi)$ is not just a subspace of $\mathbb{R}$.



 $\cong \{(\cos \theta, \sin \theta) \mid \theta \in [0, 2\pi)\}$ $\cong S^1$

모델링과 시각화

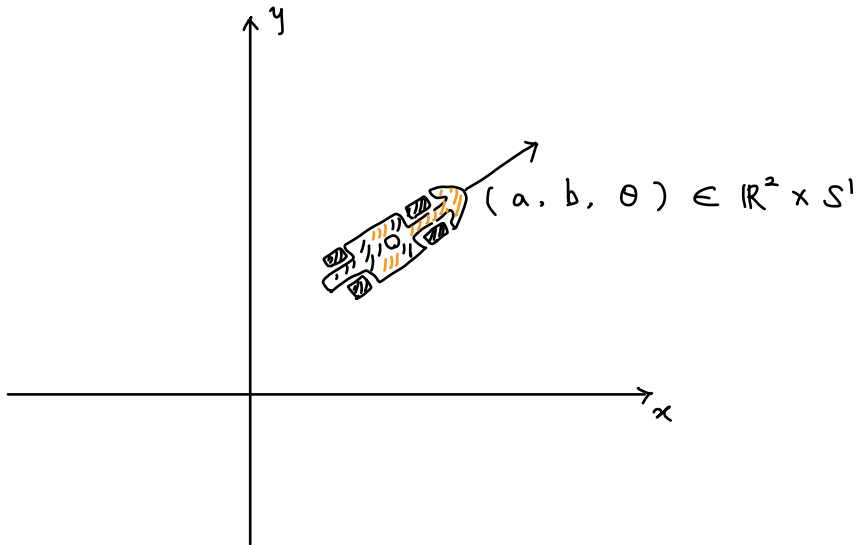


Configuration space

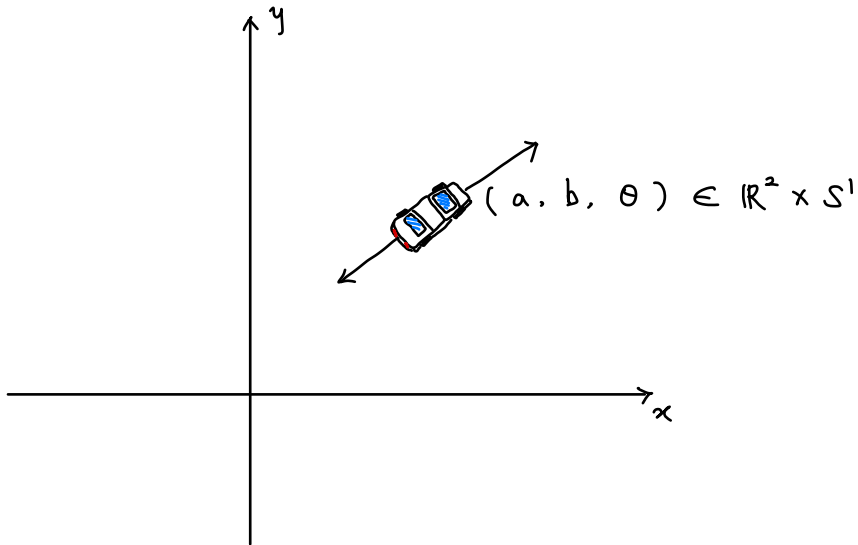


$$(a, b, \theta) \in \mathbb{R}^2 \times S^1$$

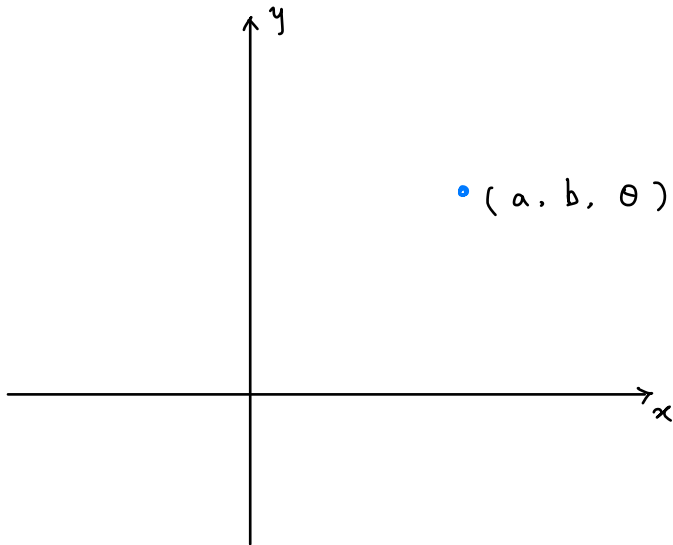
모델링과 시각화



모델링과 시각화

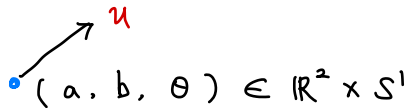


모델링과 시각화



• $(a, b, \theta) \in \mathbb{R}^2 \times S^1$

모델링과 시각화



* The velocity u of the point must satisfy $u = \|u\| (\cos \theta, \sin \theta)$

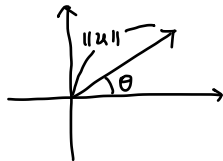
모델링과 시각화



A blue dot representing a point in a 2D plane. An arrow labeled u in red points away from the dot, representing its velocity vector.

$$(a, b, \theta) \in \mathbb{R}^2 \times S^1$$

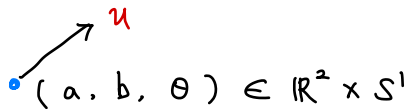
* The velocity u of the point must satisfy $u = \|u\| (\cos \theta, \sin \theta)$



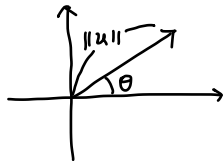
모델링과 시각화

$$u = \frac{d}{dt} (x(t), y(t))$$

$$= \left(\frac{dx}{dt}(t), \frac{dy}{dt}(t) \right)$$



* The velocity u of the point must satisfy $u = \|u\| (\cos \theta, \sin \theta)$

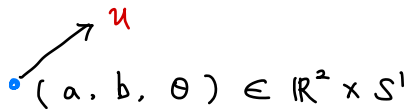


모델링과 시각화

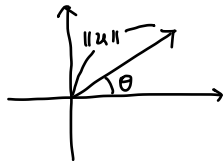
$$u = \frac{d}{dt} (x(t), y(t))$$

$$= \left(\frac{dx}{dt}(t), \frac{dy}{dt}(t) \right)$$

$$= (\|u\| \cos \theta, \|u\| \sin \theta)$$



* The velocity u of the point must satisfy $u = \|u\| (\cos \theta, \sin \theta)$



모델링과 시각화

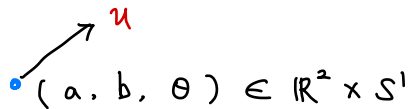
$$u = \frac{d}{dt} (x(t), y(t))$$

$$= \left(\frac{dx}{dt}(t), \frac{dy}{dt}(t) \right)$$

$$= (\|u\| \cos \theta, \|u\| \sin \theta)$$

$$\Rightarrow \dot{x} = \frac{dx}{dt}(t) = \|u\| \cos \theta,$$

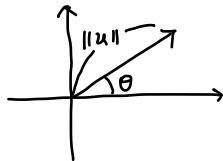
$$\dot{y} = \frac{dy}{dt}(t) = \|u\| \sin \theta.$$



A blue dot represents a point in the space $\mathbb{R}^2 \times S^1$. An arrow labeled u in red points away from the dot, representing the velocity vector.

$$(a, b, \theta) \in \mathbb{R}^2 \times S^1$$

* The velocity u of the point must satisfy $u = \|u\| (\cos \theta, \sin \theta)$



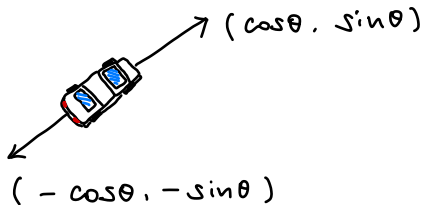
모델링과 시각화

We can control the point $\bullet$ by assigning its

speed $u_t = \overset{\text{전진}}{+1}$ or $\overset{\text{후진}}{-1}$

$$\dot{x} = u_t \cos \theta$$

$$\dot{y} = u_t \sin \theta$$



모델링과 시각화

We can control the point $\bullet$ by assigning its speed $u_t = +1$ or -1 and angular velocity $v_t = \frac{d\theta}{dt}$.

$$\dot{x} = u_t \cos \theta$$

$$\dot{y} = u_t \sin \theta$$

$$\dot{\theta} = v_t \quad \|v_t\| \leq 1$$

모델링과 시각화

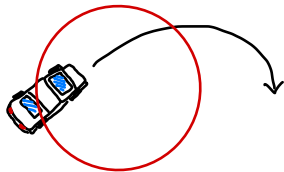
We can control the point $\bullet$ by assigning its speed $u_t = +1$ or -1 and angular velocity $v_t = \frac{d\theta}{dt}$.

$$\dot{x} = u_t \cos \theta$$

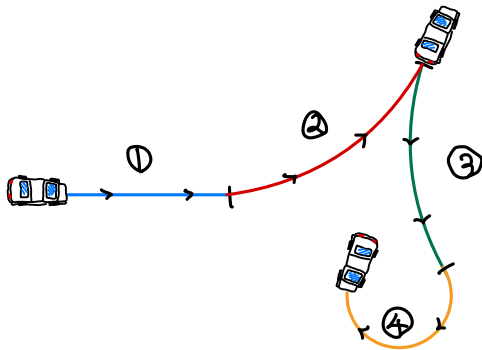
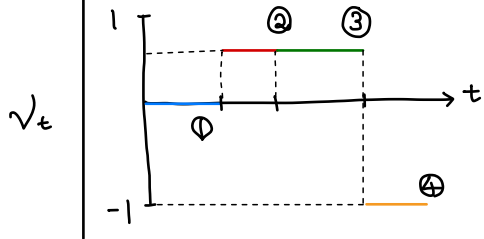
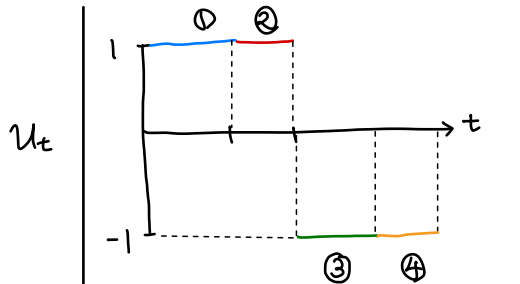
$$\dot{y} = u_t \sin \theta$$

$$\dot{\theta} = v_t$$

$$\|v_t\| \leq 1$$



Example



모델링과 시각화

We have a control system

$$\dot{x} = u \cos \theta$$

$$(x, y, \theta) \in \mathbb{R}^2 \times S^1$$

$$\dot{y} = u \sin \theta$$

$$\|u\| = 1, \quad \|v\| \leq 1$$

$$\dot{\theta} = v$$

모델링과 시각화

We have a control system

$$\dot{x} = u \cos \theta$$

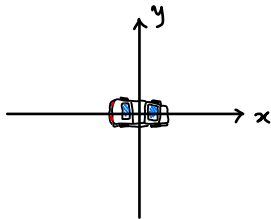
$$\dot{y} = u \sin \theta$$

$$\dot{\theta} = v$$

$$(x, y, \theta) \in \mathbb{R}^2 \times S^1$$

$$\|u\| = 1, \quad \|v\| \leq 1$$

$$f_0 = (0, 0, 0)$$



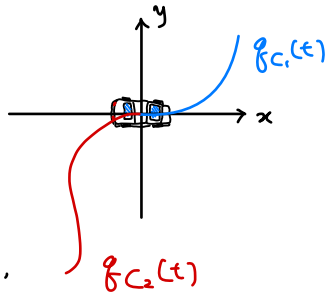
모델링과 시각화

We have a control system

$$\dot{x} = u \cos \theta \quad (x, y, \theta) \in \mathbb{R}^2 \times S^1$$

$$\dot{y} = u \sin \theta \quad \|u\| = 1, \quad \|v\| \leq 1$$

$$\dot{\theta} = v \quad q_0 = (0, 0, 0)$$



For each control parameter $c = \begin{pmatrix} u \\ v \end{pmatrix}$,

we obtain a trajectory $f_c(t)$

모델링과 시각화

Definition

The set $A_{f_0} = \{ f_c(t) \mid f_u(0) = f_0,$

c is a control parameter,

$t \geq 0$

is called the **attainable set** of a control system.

모델링과 시각화

Definition

The set $A_{f_0} = \{ f_c(t) \mid f_u(0) = f_0,$

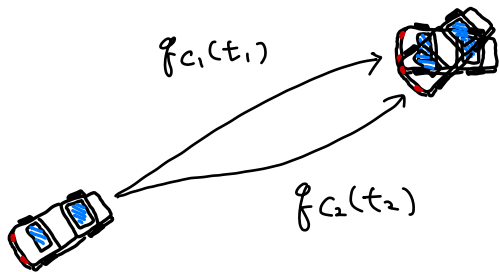
c is a control parameter,

$t \geq 0$

is called the **attainable set** of a control system.

Q. $A_{f_0} = \mathbb{R}^2 \times S^1$?

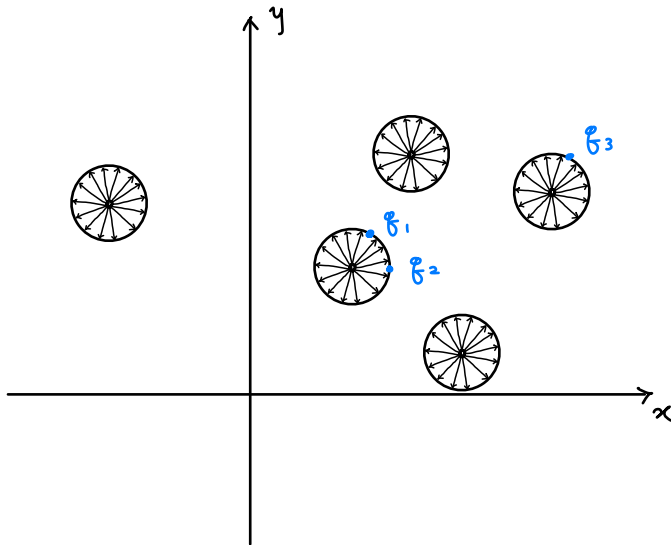
모델링과 시각화



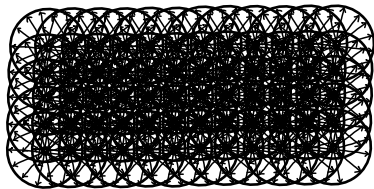
They are not in the same point in $\mathbb{R}^2 \times S^1$.

((x, y, θ) 에서 θ 가 $\frac{2\pi}{2}$)

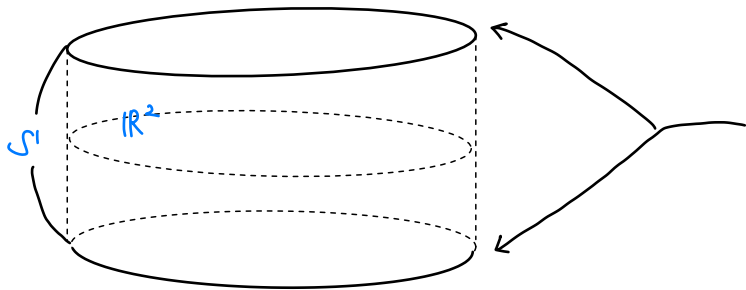
$\mathbb{R}^2 \times \mathcal{S}^1$ 시각화



$\mathbb{R}^2 \times \mathcal{U}'$ 시각화

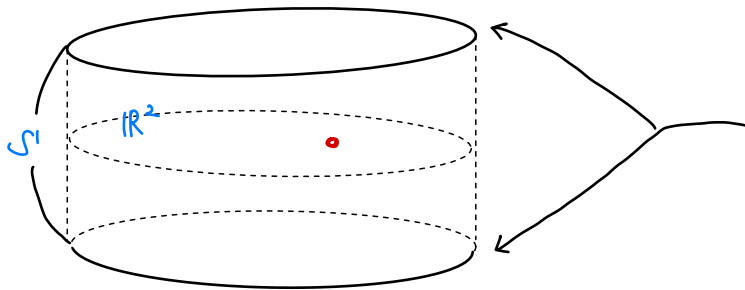


$\mathbb{R}^2 \times S^1$ 시각화



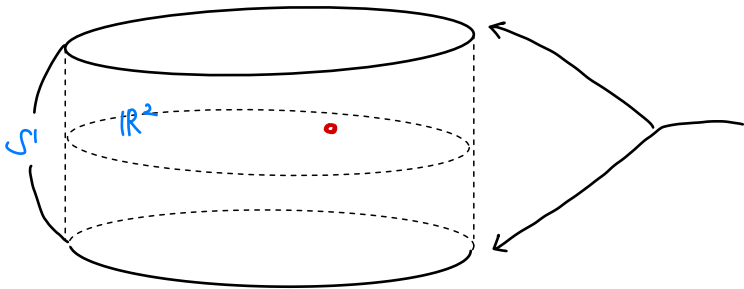
윗면이랑
아랫면이
이어져요.

$\mathbb{R}^2 \times S^1$ 시각화



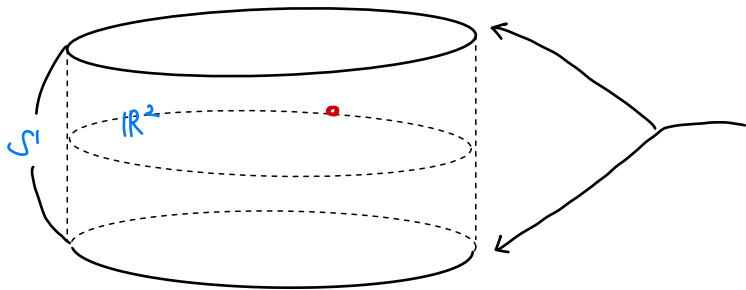
윗면이랑
아랫면이
이어져요.

$\mathbb{R}^2 \times S^1$ 시각화



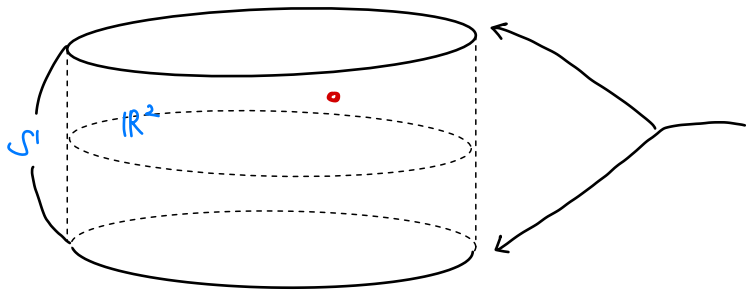
윗면이랑
아랫면이
이어져요.

$\mathbb{R}^2 \times S^1$ 시각화



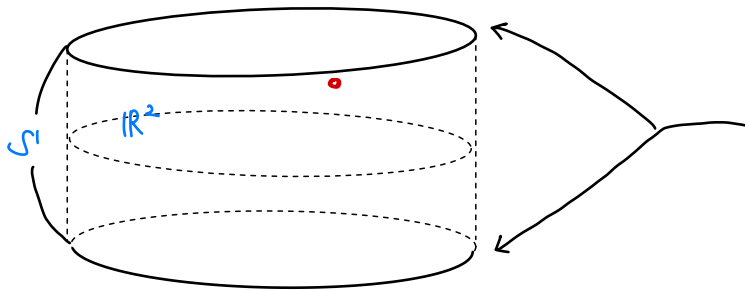
윗면이랑
아랫면이
이어져요.

$\mathbb{R}^2 \times S^1$ 시각화



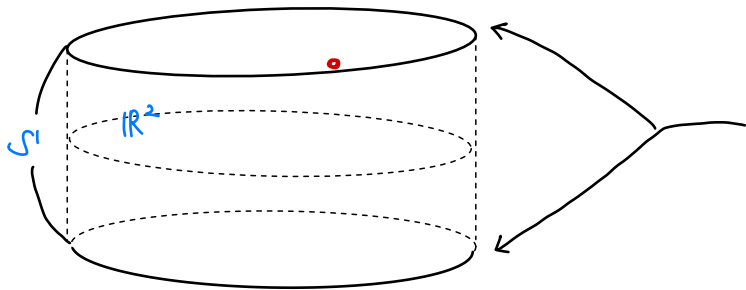
윗면이랑
아랫면이
이어져요.

$\mathbb{R}^2 \times S^1$ 시각화



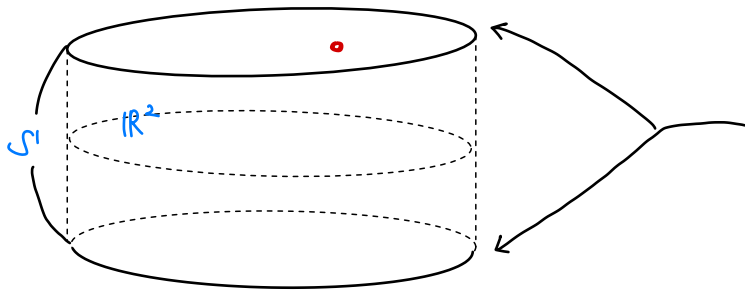
윗면이랑
아랫면이
이어져요.

$\mathbb{R}^2 \times S^1$ 시각화



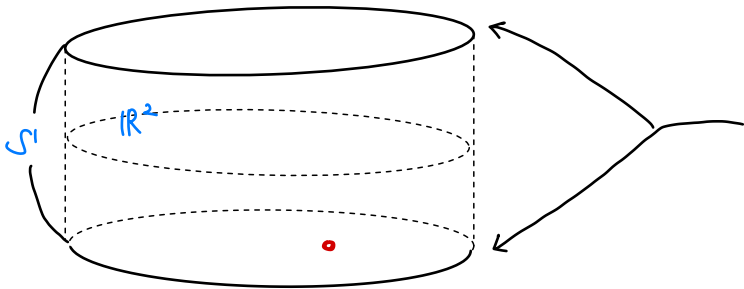
윗면이랑
아랫면이
이어져요.

$\mathbb{R}^2 \times S^1$ 시각화



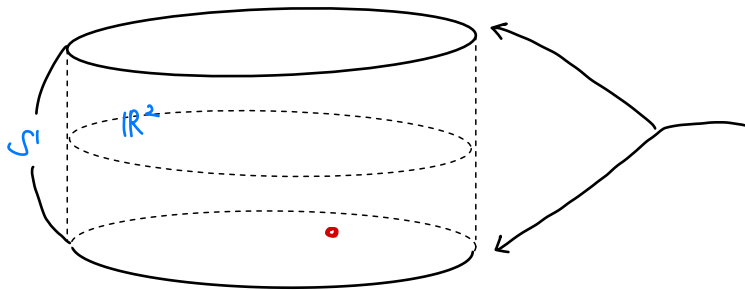
윗면이랑
아랫면이
이어져요.

$\mathbb{R}^2 \times S^1$ 시각화



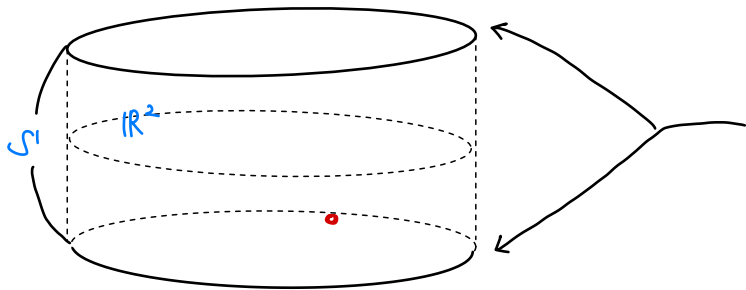
윗면이랑
아랫면이
이어져요.

$\mathbb{R}^2 \times S^1$ 시각화



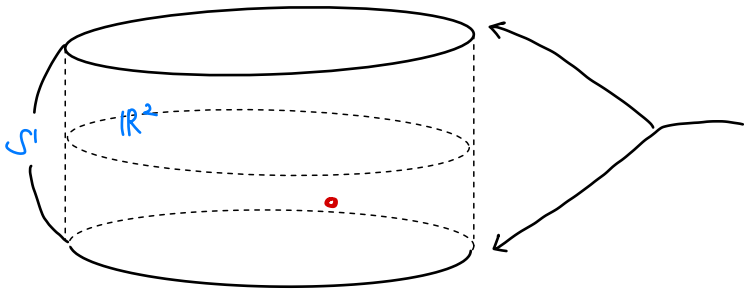
윗면이랑
아랫면이
이어져요.

$\mathbb{R}^2 \times S^1$ 시각화



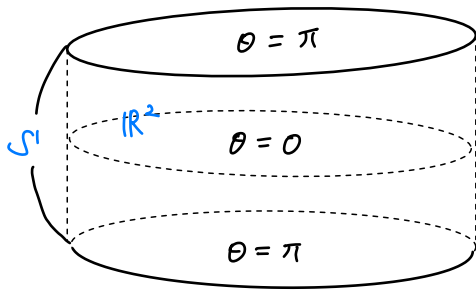
윗면이랑
아랫면이
이어져요.

$\mathbb{R}^2 \times S^1$ 시각화



윗면이랑
아랫면이
이어져요.

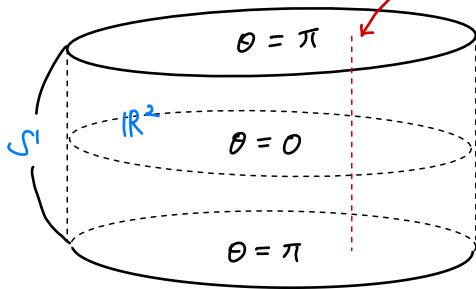
$\mathbb{R}^2 \times S^1$ 시각화



층 하나가
방향을
나타냄.

$\mathbb{R}^2 \times S^1$ 시각화

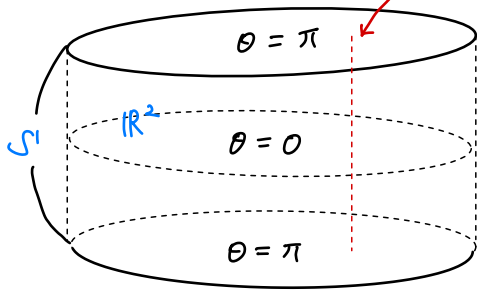
점선을 따라 $\mathbb{R}^2$ 에서는 같은
위치지만 방향이 다름



층 하나가
방향을
나타내요.

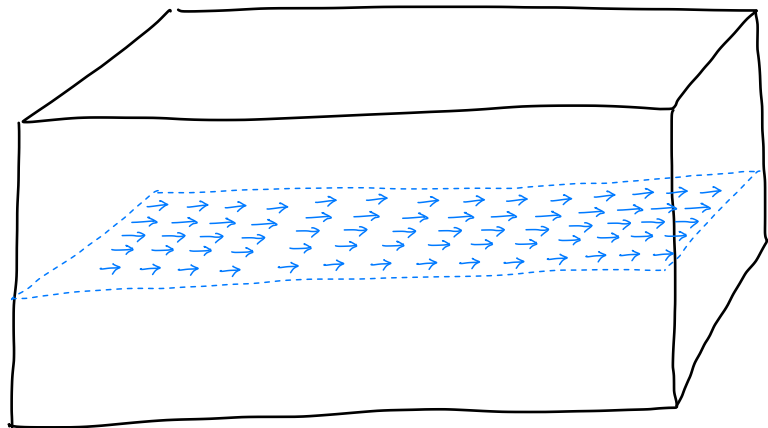
$\mathbb{R}^2 \times S^1$ 시각화

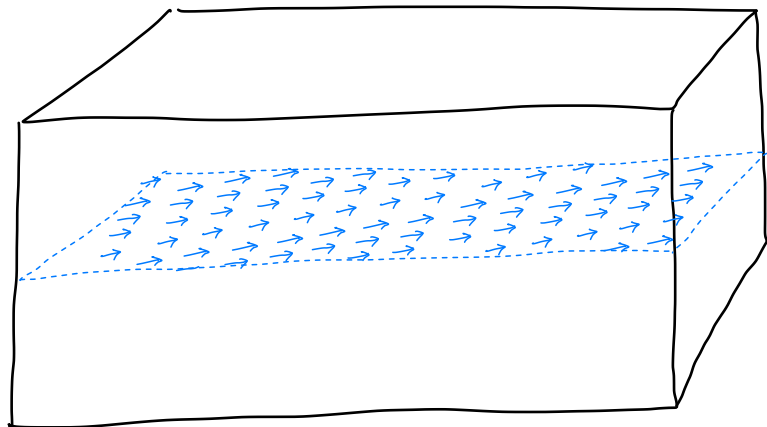
점선을 따라 $\mathbb{R}^2$ 에서는 같은
위치지만 방향이 다름

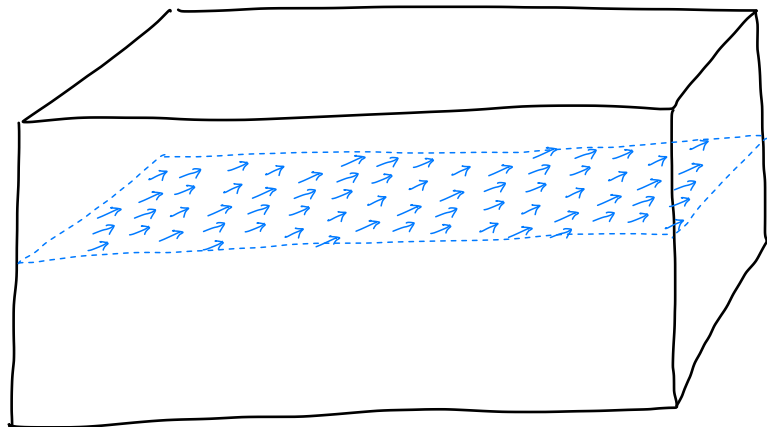


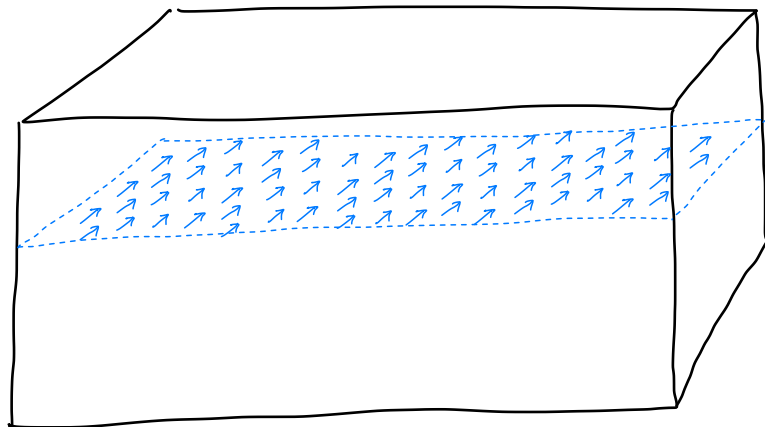
층 하나가
방향을
나타내요.

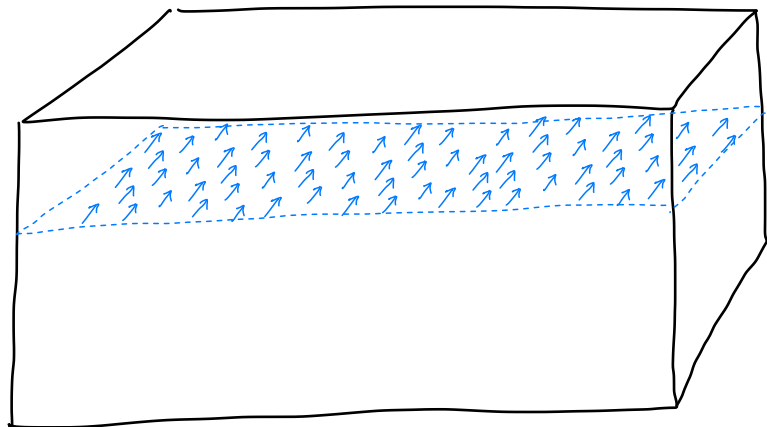
각 층이 나타내는 방향을 화살표로 나타내면 ...



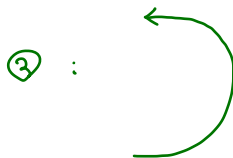
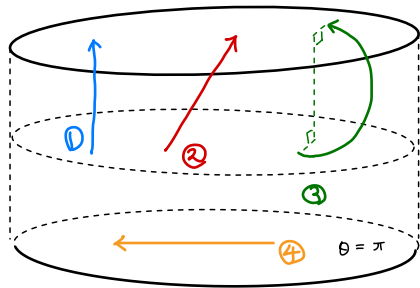




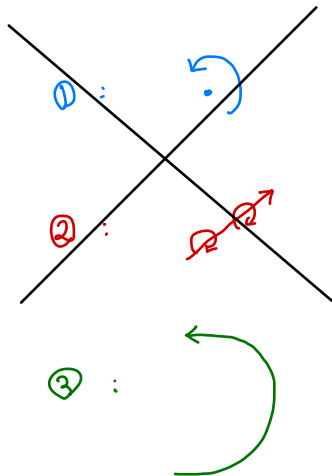
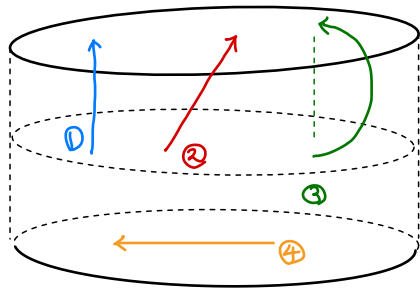




$\mathbb{R}^2 \times S^1$ 시각화



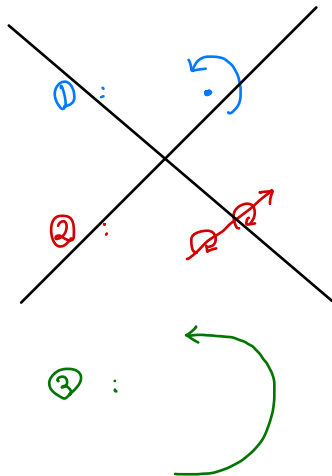
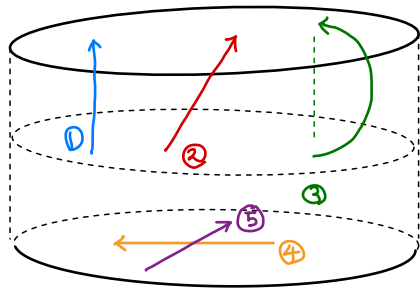
$\mathbb{R}^2 \times S^1$ 시각화



Impossible.



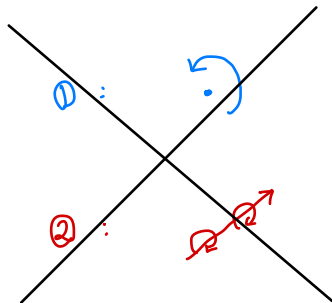
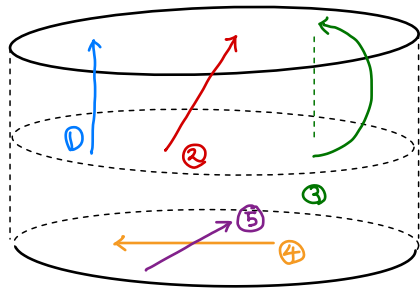
Quiz



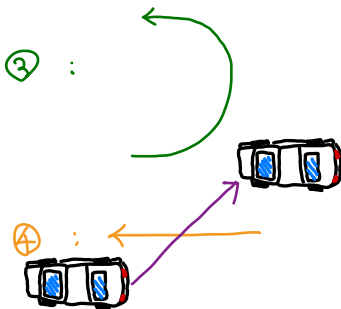
Impossible.



Quiz



Impossible.



Recall

We have a control system

$$\dot{x} = u \cos \theta \quad (x, y, \theta) \in \mathbb{R}^2 \times S^1$$

$$\dot{y} = u \sin \theta \quad \|u\| = 1, \quad \|v\| \leq 1$$

$$\dot{\theta} = v \quad f_0 = (0, 0, 0),$$

and the attainable set

$$A_{f_0} = \{ f_c(t) \mid f_u(0) = f_0, \quad c \text{ is a control parameter, } t \geq 0 \}.$$

Recall

We have a control system

$$\dot{x} = u \cos \theta$$

$$\dot{y} = u \sin \theta$$

$$\dot{\theta} = v$$

$$(x, y, \theta) \in \mathbb{R}^2 \times S^1$$

$$\|u\| = 1, \quad \|v\| \leq 1$$

$$f_0 = (0, 0, 0),$$

자동차는
남작해도
3차원에
살고있다.

and the attainable set

$$A_{f_0} = \{ f_c(t) \mid f_u(0) = f_0, \quad c \text{ is a control parameter, } t \geq 0 \}.$$

2. Tangent space and Rashevskii - Chow Theorem

Tangent space

자동차

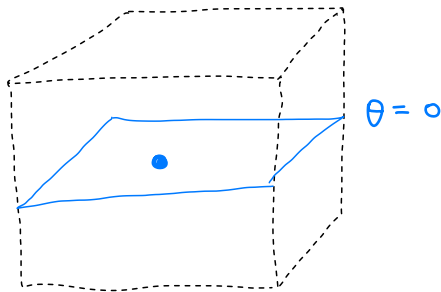
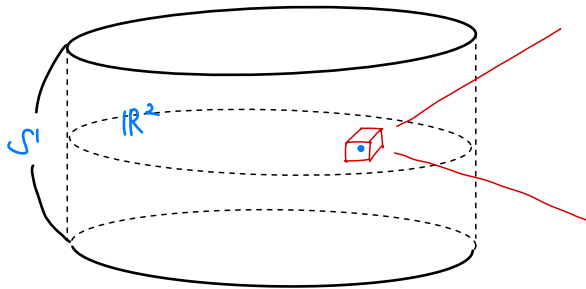
vs

마우스



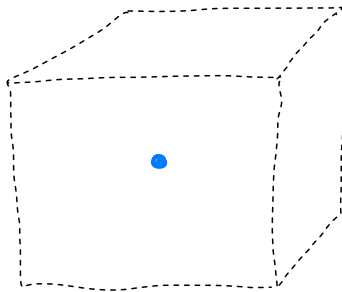
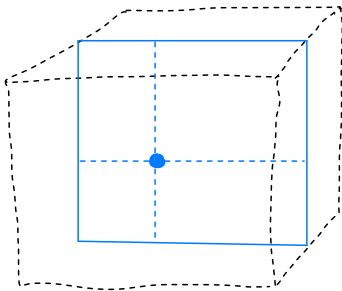
Tangent space

자동차 vs 마우스



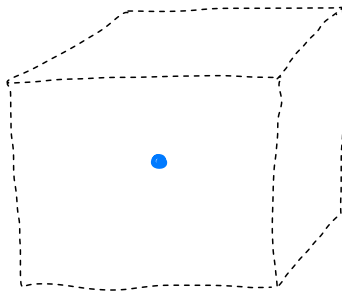
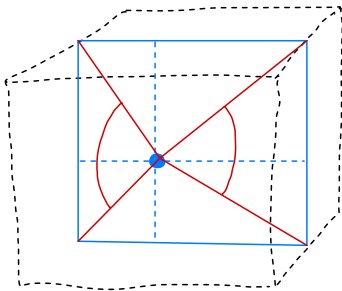
Tangent space

자동차 vs 마우스



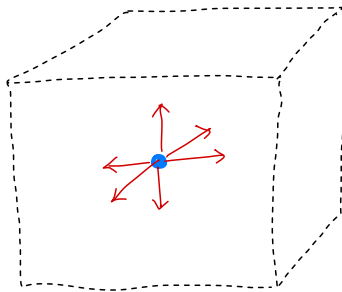
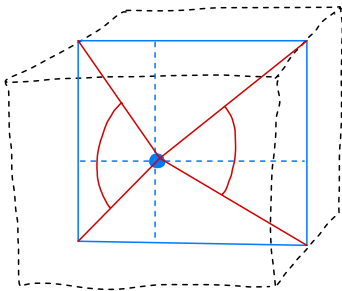
Tangent space

자동차 vs 마우스



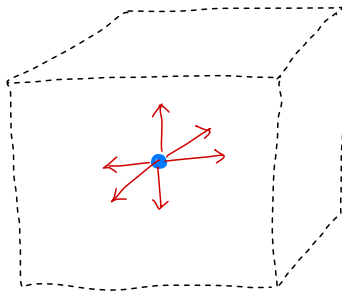
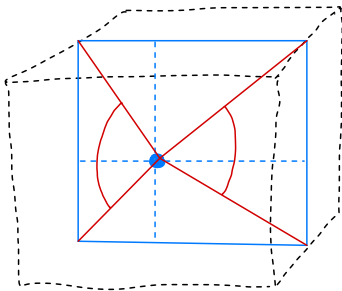
Tangent space

자동차 vs 마우스



Tangent space

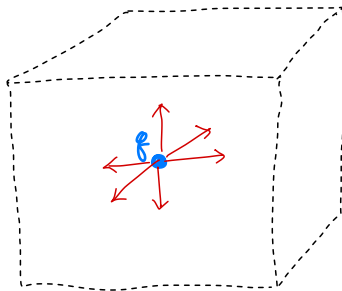
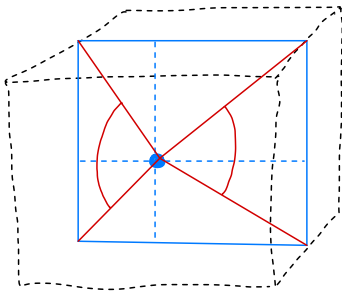
자동차 vs 마우스



Tangent space of $\mathbb{R}^2 \times S^1 : \mathbb{R}^3$

Tangent space

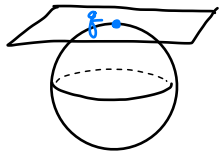
자동차 vs 마우스



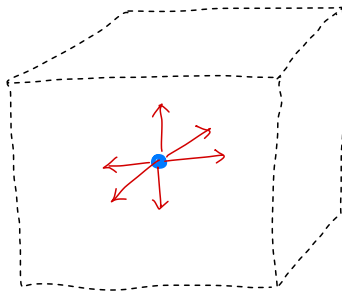
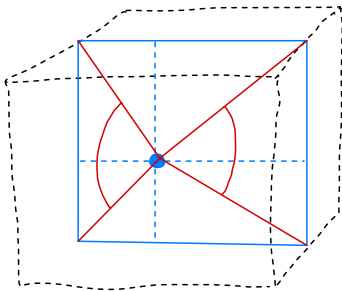
$$T_q(\mathbb{R}^2 \times S^1) \cong \mathbb{R}^3$$

Tangent space

$$\text{ex) } T_f S^2 \cong \mathbb{R}^2$$



자동차 vs 마우스



$$T_f(\mathbb{R}^2 \times S^1) \cong \mathbb{R}^3$$

자동차의 궤

궤 : $u(\rightarrow)$ 와 $v(\hookrightarrow)$ 를 조합하면,

u 에 수직한 방향 ($\uparrow$)을 만들 수

있지 않을까?

자등차의 η

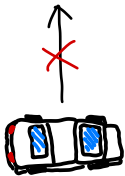
η : $u(\rightarrow)$ 와 $v(\hookrightarrow)$ 를 조합하면,

u 에 수직인 방향 ($\uparrow$)을 만들 수

있지 않을까?

그러면 $\text{span} \{ \rightarrow, \uparrow, \hookrightarrow \} = \mathbb{R}^3$ 일 텐 데 !

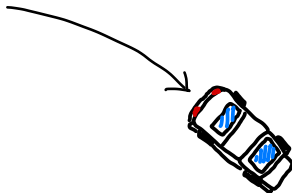
Example



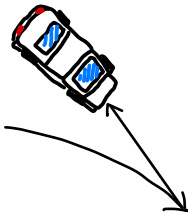
Example



Example



Example



Example



Example



$$\text{span} \{ \rightarrow, \uparrow, \curvearrowright \} = \mathbb{R}^3$$

completely
non-holonomic

(만들어 낸 방향까지 해서
tangent space $\frac{2}{2}$ 차원)

Rashevskii - Chow theorem

Theorem (Rashevskii - Chow)

- ① completely nonholonomic
- ② M is connected

$$\Rightarrow O_f = M, \quad \forall f \in M$$

In our case, $A_f = O_f$.

Rashevskii - Chow theorem

Theorem (Rashevskii - Chow)

① completely nonholonomic

~~② M is connected~~ ($\mathbb{R}^2 \times S^1$ is connected)

$$\Rightarrow \textcircled{0}_f = M, \quad \forall f \in M \quad A_f = \mathbb{R}^2 \times S^1, \quad \forall f \in \mathbb{R}^2 \times S^1$$

Rashevskii - Chow theorem

Theorem (Rashevskii - Chow)

① completely nonholonomic

② ~~M is connected~~ ($\mathbb{R}^2 \times S^1$ is connected)

$$\Rightarrow \textcircled{O}_f = \textcircled{M}, \quad \forall f \in \textcircled{M} \quad A_f = \mathbb{R}^2 \times S^1, \quad \forall f \in \mathbb{R}^2 \times S^1$$

자 동 차 는 제 약 을 넘 어 자 유 롭 고 싶 다.

3. Orbit of Lie Bracket

Example



Orbit 과 Lie Bracket

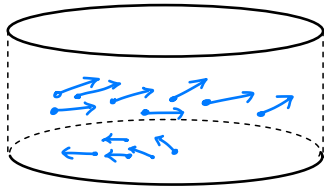
Control 을 어떻게 할 것인가 ... ?

Orbit 과 Lie Bracket

Control 을 어떻게 합성하지 ... ?

Vector field 로 생각하자 !

$$\mathbb{R}^2 \times S^1 \longrightarrow \mathbb{R}^3$$

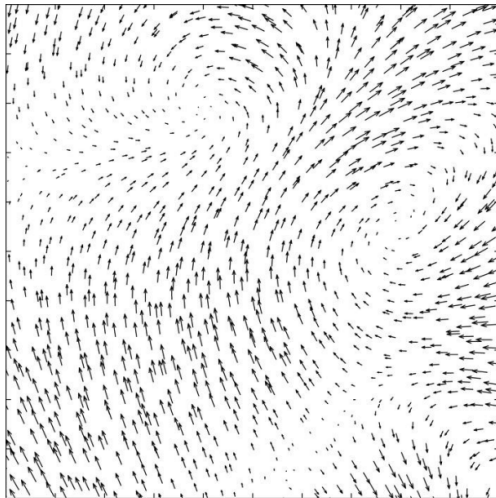


Vector field

Example 1

$$V: \mathbb{R}^2 \longrightarrow T\mathbb{R}^2 = \mathbb{R}^2$$

각 점에 벡터를 대응시킴.

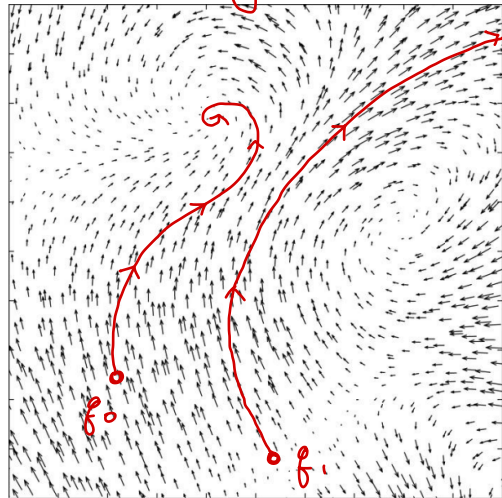


Vector field

Example)

$$V: \mathbb{R}^2 \longrightarrow T\mathbb{R}^2 = \mathbb{R}^2$$

각 점에 벡터를 대응시킴.

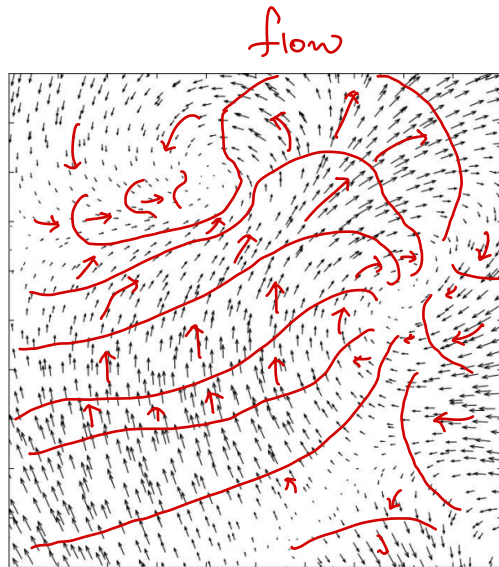


Vector field

Example)

$$V: \mathbb{R}^2 \rightarrow T\mathbb{R}^2 = \mathbb{R}^2$$

각 점에 벡터를 대응시킴.

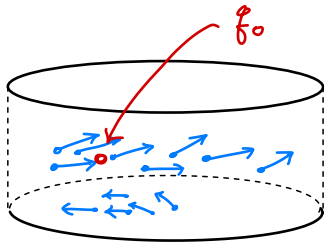


Orbit 과 Lie Bracket

Control 을 어떻게 합성하지 ... ?

Vector field 로 생각하자 !

$$\mathbb{R}^2 \times S^1 \longrightarrow \mathbb{R}^3$$

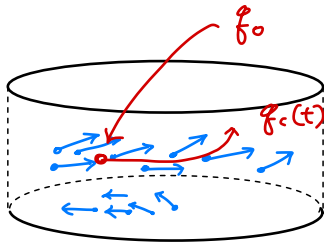


Orbit 과 Lie Bracket

Control 을 어떻게 합성하지 ... ?

Vector field 로 생각하자 !

$$\mathbb{R}^2 \times S^1 \longrightarrow \mathbb{R}^3$$

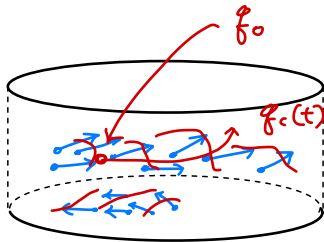


Orbit 과 Lie Bracket

Control 을 어떻게 합성하지 ... ?

Vector field 로 생각하자 !

$$\mathbb{R}^2 \times S^1 \longrightarrow \mathbb{R}^3$$



Orbit at Lie Bracket

$$\dot{x} = u \cos \theta \quad (x, y, \theta) \in \mathbb{R}^2 \times S^1$$

$$\dot{y} = u \sin \theta \quad \|u\| = 1, \quad \|v\| \leq 1$$

$$\dot{\theta} = v \quad f_0 = (0, 0, 0)$$

$$\text{Let } q = (x, y, \theta).$$

$$\dot{q} = (\dot{x}, \dot{y}, \dot{\theta}) = (u \cos \theta, u \sin \theta, v)$$

Orbit at Lie Bracket

$$\text{Let } q = (x, y, \theta).$$

$$\dot{q} = (\dot{x}, \dot{y}, \dot{\theta}) = (u \cos \theta, u \sin \theta, v)$$

$$\text{Define } f_c : \mathbb{R}^2 \times S^1 \longrightarrow \mathbb{R}^3$$

$$q \longmapsto \dot{q} = (u \cos \theta, u \sin \theta, v)$$

Orbit 과 Lie Bracket

$$\text{Let } q = (x, y, \theta).$$

$$\dot{q} = (\dot{x}, \dot{y}, \dot{\theta}) = (u \cos \theta, u \sin \theta, v)$$

$$\text{Define } f_c : \mathbb{R}^2 \times S^1 \longrightarrow \mathbb{R}^3$$

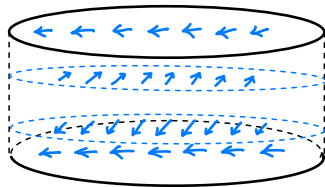
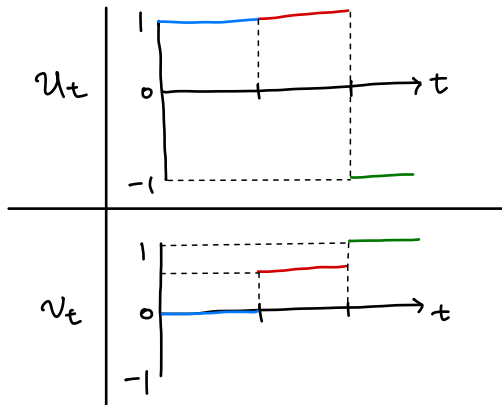
$$q \longmapsto \dot{q} = (u \cos \theta, u \sin \theta, v)$$

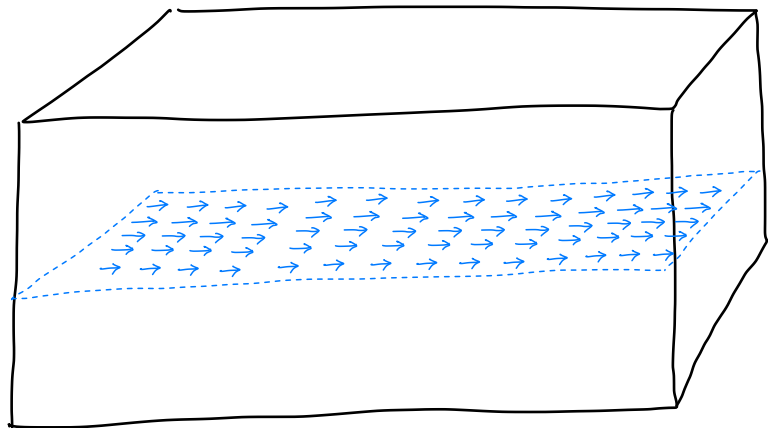
각 점에서의 속도를 정해준다고 생각하면 됩니다.

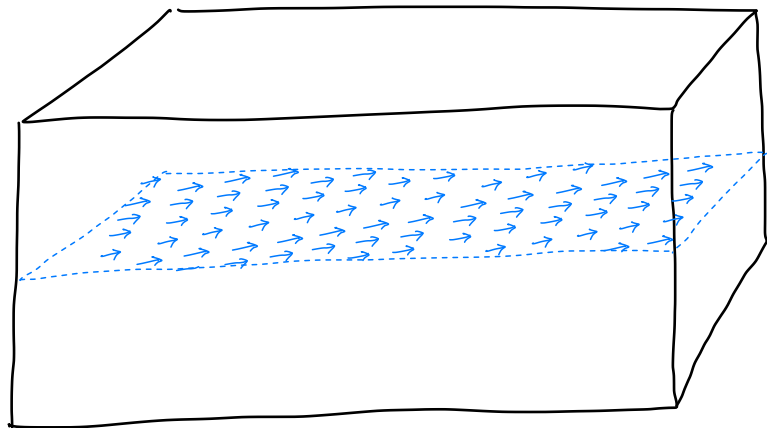
Example

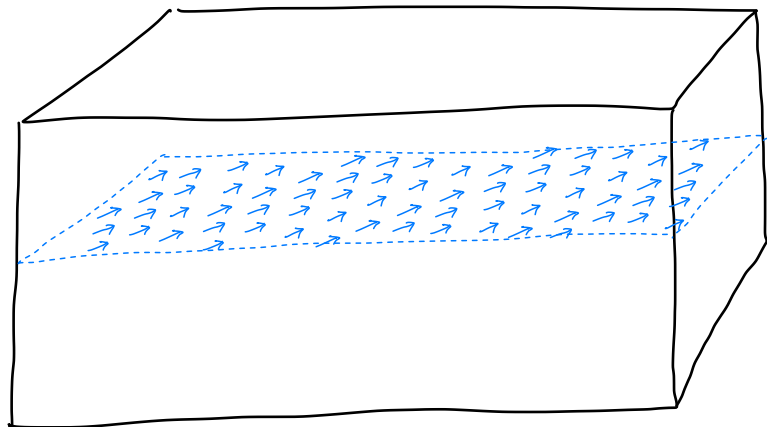
Define $f_c : \mathbb{R}^2 \times S^1 \longrightarrow \mathbb{R}^3$

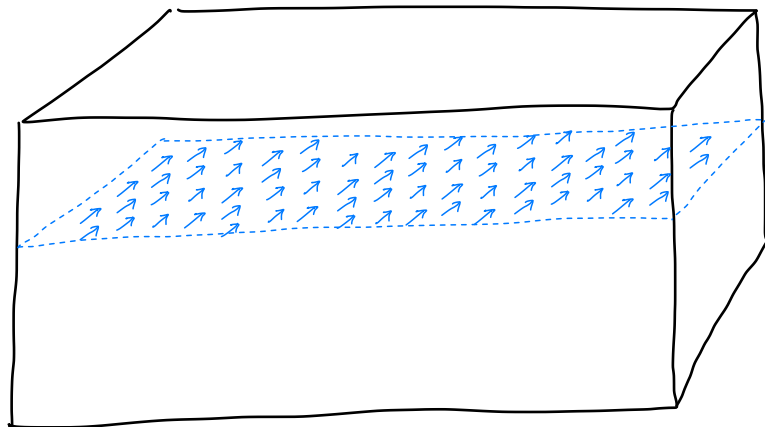
$$q \longmapsto \dot{q} = (u_t \cos \theta, u_t \sin \theta, v_t)$$

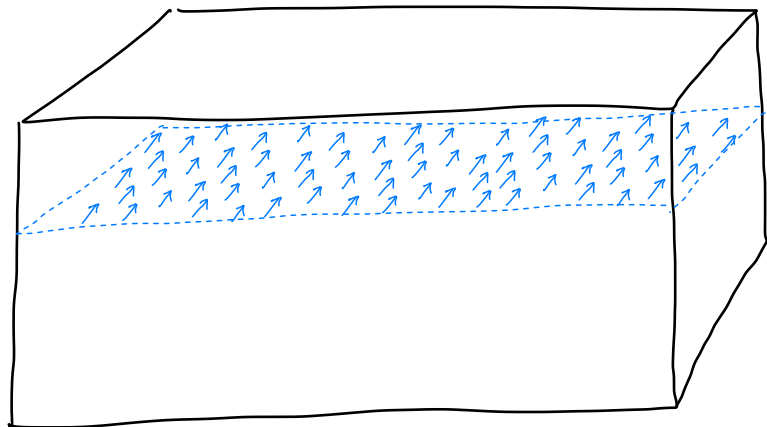








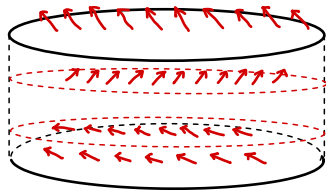
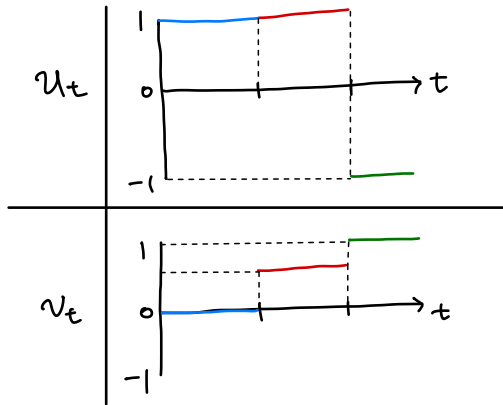




Example

Define $f_c : \mathbb{R}^2 \times S^1 \longrightarrow \mathbb{R}^3$

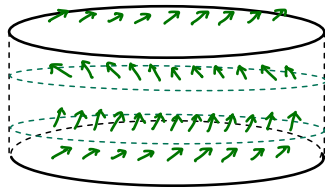
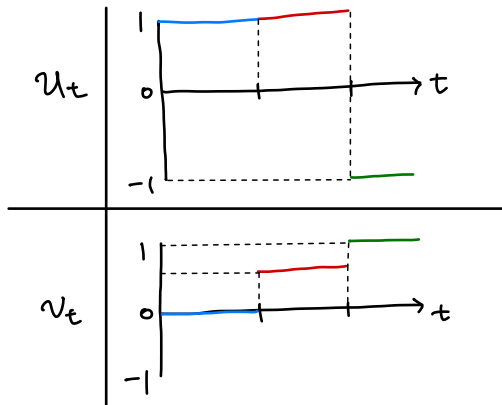
$$q \longmapsto \dot{q} = (u_t \cos \theta, u_t \sin \theta, v_t)$$



Example

Define $f_c : \mathbb{R}^2 \times S^1 \longrightarrow \mathbb{R}^3$

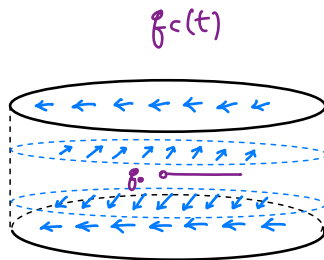
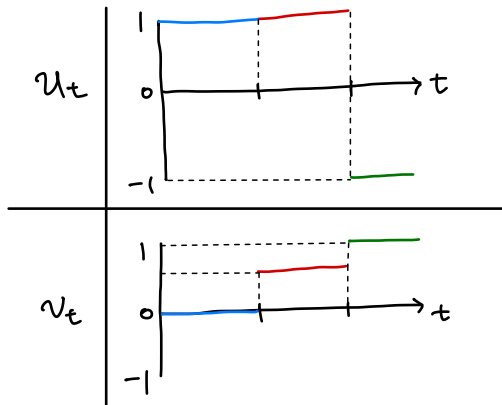
$$q \longmapsto \dot{q} = (u_t \cos \theta, u_t \sin \theta, v_t)$$



Example

Define $f_c : \mathbb{R}^2 \times S^1 \longrightarrow \mathbb{R}^3$

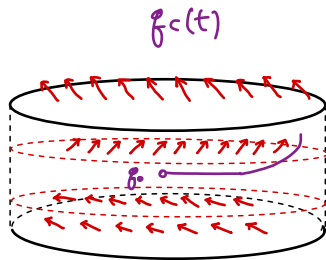
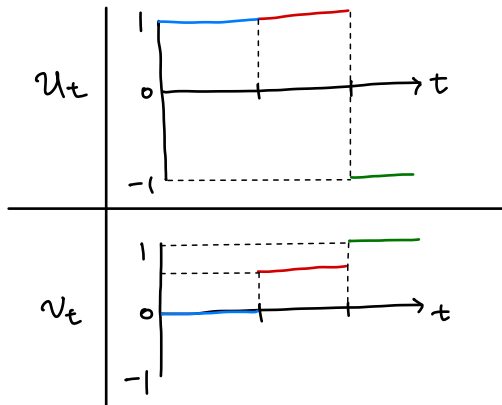
$$q \longmapsto \dot{q} = (u_t \cos \theta, u_t \sin \theta, v_t)$$



Example

Define $f_c : \mathbb{R}^2 \times S^1 \longrightarrow \mathbb{R}^3$

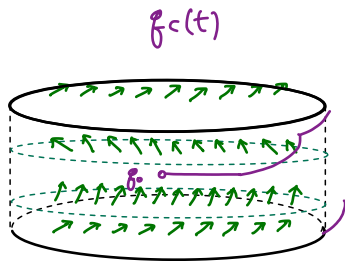
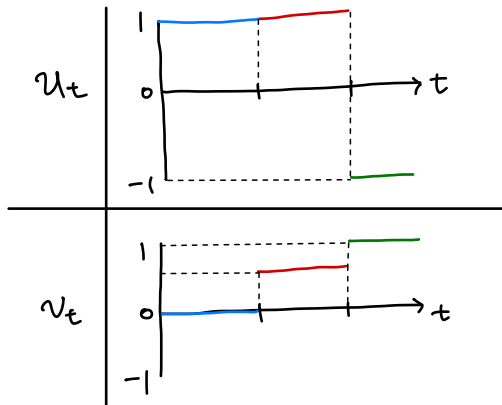
$$q \longmapsto \dot{q} = (u_t \cos \theta, u_t \sin \theta, v_t)$$



Example

Define $f_c : \mathbb{R}^2 \times S^1 \longrightarrow \mathbb{R}^3$

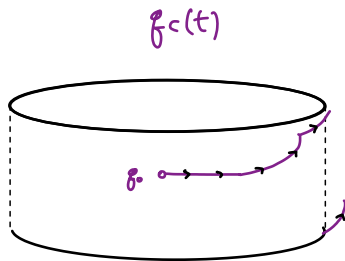
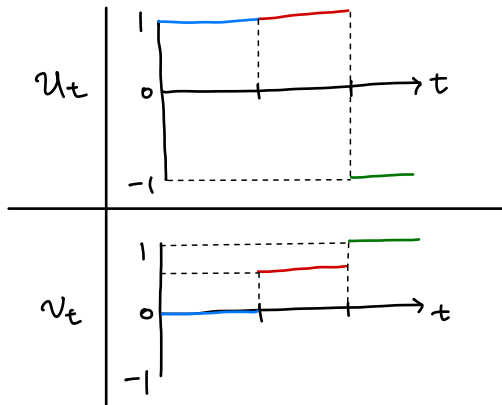
$$q \longmapsto \dot{q} = (u_t \cos \theta, u_t \sin \theta, v_t)$$



Example

Define $f_c : \mathbb{R}^2 \times S^1 \longrightarrow \mathbb{R}^3$

$$q \longmapsto \dot{q} = (u_t \cos \theta, u_t \sin \theta, v_t)$$



Orbit at Lie Bracket

Does the vector field have to be time-dependent?

$$f_c : T \times \mathbb{R}^2 \times S^1 \longrightarrow \mathbb{R}^3$$

Orbit at Lie Bracket

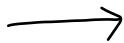
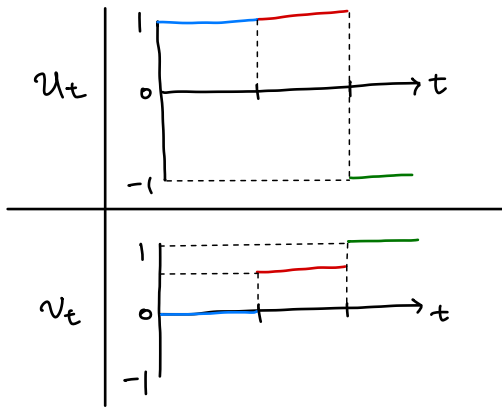
Does the vector field have to be time-dependent?

$$f_c : T \times \mathbb{R}^2 \times S^1 \longrightarrow \mathbb{R}^3$$

No, not necessarily.

Orbit at Lie Bracket

We can work with a family of vector fields.



$$\left\{ \begin{array}{c} \text{blue } F \\ \text{blue } F \end{array}, \begin{array}{c} \text{red } F \\ \text{red } F \end{array}, \begin{array}{c} \text{green } F \\ \text{green } F \end{array} \right\}$$

Orbit & Lie Bracket

Let $\mathcal{F} \subseteq \text{Vect}(\mathbb{R}^2 \times S')$

be the family of vector fields corresponding to the controls.

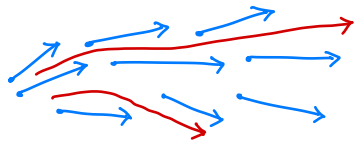
Orbit & Lie Bracket

Definition (for trajectories)

An **integral curve** of $f \in \mathcal{F}$

is a differentiable curve $\theta: \mathbb{R} \rightarrow \mathbb{R}^2 \times S^1$

s.t. $\theta'(t) = f(\theta(t))$, $\forall t \in \mathbb{R}$



Orbit & Lie Bracket

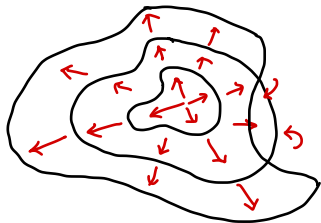
Theorem (for trajectories)

Let $f \in \mathcal{F}$.

$\exists$ global flow Θ s.t.

s.t. $\Theta(-, \xi) : \mathbb{R} \rightarrow \mathbb{R}^2 \times S^1$

is an integral curve of f .



Orbit & Lie Bracket

Theorem (for trajectories)

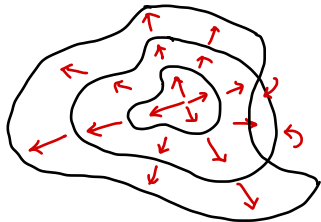
Let $f \in \mathcal{F}$.

$\exists$ global flow Θ s.t.

s.t. $\Theta(-, \xi) : \mathbb{R} \rightarrow \mathbb{R}^2 \times S^1$

is an integral curve of f .

(Fundamental thm. on flows
+ f is left invariant under
a Lie group $(SO(2) \times \mathbb{R}^2)$)



Orbit at Lie Bracket

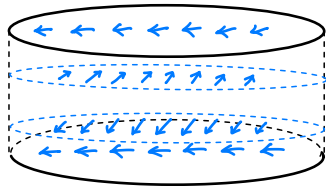
Denote by θ_f the corresponding flow of $f \in \mathcal{F}$.

Orbit at Lie Bracket

Denote by θ_f the corresponding flow of $f \in \mathcal{F}$.

Example: If $f(p) = (\cos \theta, \sin \theta, 0)$,

$f(p):$

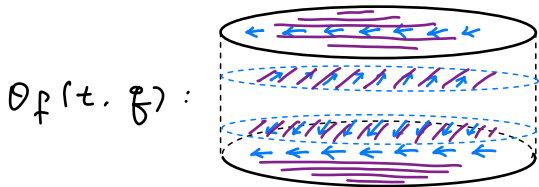
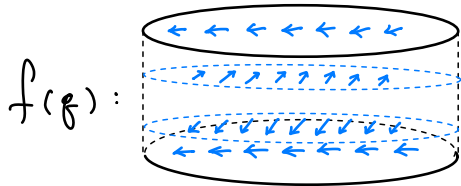


Orbit of Lie Bracket

Denote by θ_f the corresponding flow of $f \in \mathcal{F}$.

Example: If $f(p) = (\cos \theta, \sin \theta, 0)$,

$$\begin{aligned}\text{then } \theta_f(t, p) &= p + (\cos \theta t, \sin \theta t, 0) \\ &= (x + \cos \theta t, y + \sin \theta t, 0)\end{aligned}$$



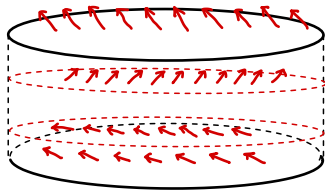
Orbit at Lie Bracket

Denote by θ_f the corresponding flow of $f \in \mathcal{F}$.

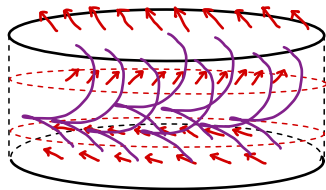
Example: If $f(p) = (\cos \theta, \sin \theta, 1)$,

then $\theta_f(t, p) = \dots$

$f(p):$



$\theta_f(t, p):$



Orbit 과 Lie Bracket

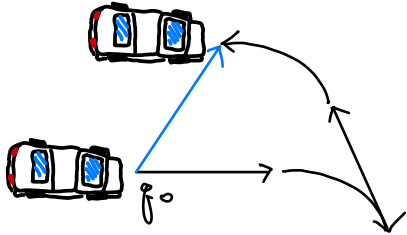
이제 $f(\theta) = (\cos\theta, \sin\theta, 0)$ 과

$g(\theta) = (\cos\theta, \sin\theta, 1)$ 의 조합으로

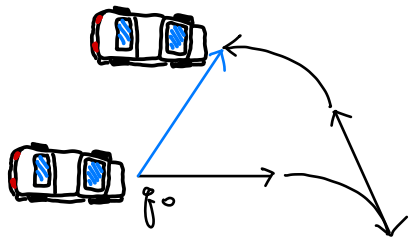
새로운 방향을 만들어 봅시다.

연산을 어떻게 해야 할까요?

Lie bracket

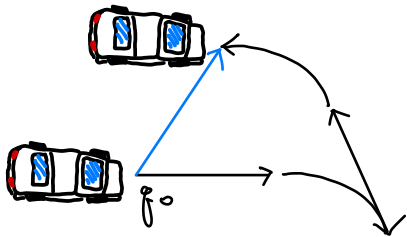


Lie bracket



$$= \theta_g(-t, \theta_f(-t, \theta_g(t, \theta_f(t, f_0)))) - f_0$$

Lie bracket



$$= \theta_g(-t, \theta_f(-t, \theta_g(t, \theta_f(t, f_0)))) - f_0$$

$$= \theta_g^{(-t)} \circ \theta_f^{(-t)} \circ \theta_g^{(t)} \circ \theta_f^{(t)}(f_0) - f_0$$

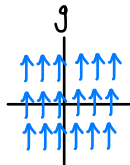
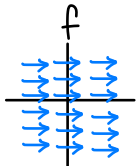
Lie bracket

$$[f, g] = \left. \frac{d}{dt} \right|_{t=0} \theta_g^{(-\sqrt{t})} \circ \theta_f^{(-\sqrt{t})} \circ \theta_g^{(\sqrt{t})} \circ \theta_g^{(\sqrt{t})} (f_0)$$

Quiz

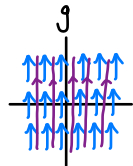
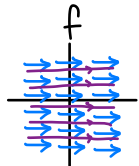
On $\mathbb{R}^2$,

Let $f(x) = (1, 0)$, $g(x) = (0, 1)$.



Quiz

On $\mathbb{R}^2$,

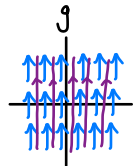
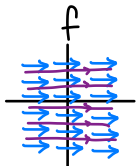


Let $f(x) = (1, 0)$, $g(x) = (0, 1)$.

Then $\theta_f^{(t)}(x, y) = (x + t, y)$, $\theta_g^{(t)}(x, y) = (x, y + t)$.

Quiz

On $\mathbb{R}^2$,



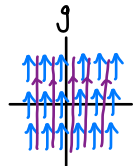
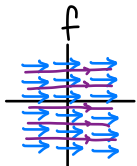
Let $f(p) = (1, 0)$, $g(p) = (0, 1)$.

Then $\theta_f^{(t)}(x, y) = (x+t, y)$, $\theta_g^{(t)}(x, y) = (x, y+t)$.

Compute $[f, g] = \frac{d}{dt}\bigg|_{t=0} \theta_g^{(-\sqrt{t})} \circ \theta_f^{(-\sqrt{t})} \circ \theta_g^{(\sqrt{t})} \circ \theta_f^{(\sqrt{t})}(p_0)$.

Quiz

On $\mathbb{R}^2$,



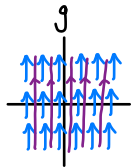
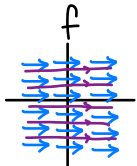
Let $f(p) = (1, 0)$, $g(p) = (0, 1)$.

Then $\theta_f^{(t)}(x, y) = (x+t, y)$, $\theta_g^{(t)}(x, y) = (x, y+t)$.

Compute $[f, g] = \frac{d}{dt} \Big|_{t=0} \theta_g^{(-\sqrt{t})} \circ \theta_f^{(-\sqrt{t})} \circ \theta_g^{(\sqrt{t})} \circ \theta_f^{(\sqrt{t})}(p_0)$.

Quiz

On $\mathbb{R}^2$,

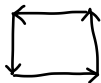


Let $f(p) = (1, 0)$, $g(p) = (0, 1)$.

Then $\theta_f^{(t)}(x, y) = (x+t, y)$, $\theta_g^{(t)}(x, y) = (x, y+t)$.

Compute $[f, g] = \frac{d}{dt}\bigg|_{t=0} \theta_g^{(-\sqrt{t})} \circ \theta_f^{(-\sqrt{t})} \circ \theta_g^{(\sqrt{t})} \circ \theta_f^{(\sqrt{t})}(p_0)$.

$$= (0, 0)$$



Lie bracket

$$[f, g] = \left. \frac{d}{dt} \right|_{t=0} \theta_g^{(-\sqrt{t})} \circ \theta_f^{(-\sqrt{t})} \circ \theta_g^{(\sqrt{t})} \circ \theta_g^{(\sqrt{t})} (f_0)$$

계산 어떻게 할건데 ...

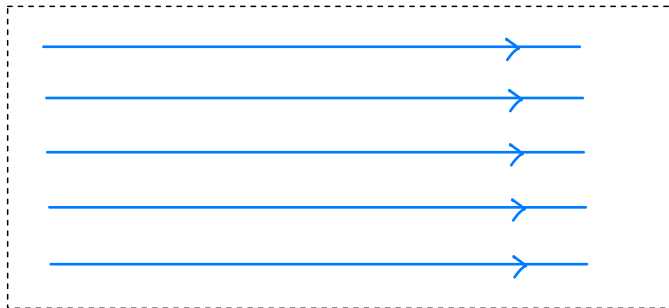
Lie derivative

Define a vector field $L_{\mathbf{f}} \mathbf{g} : \mathbb{R}^2 \times S^1 \rightarrow \mathbb{R}^3$ by

$$(L_{\mathbf{f}} \mathbf{g})(\mathbf{p}) = \left. \frac{d}{dt} \right|_{t=0} \theta_{\mathbf{f}*}^{(-t)}(\theta_{\mathbf{f}}^{(t)} \mathbf{p}) [\mathbf{g}(\theta_{\mathbf{f}}^{(t)}(\mathbf{p}))]$$

Lie derivative

θf



Lie derivative

vector field g



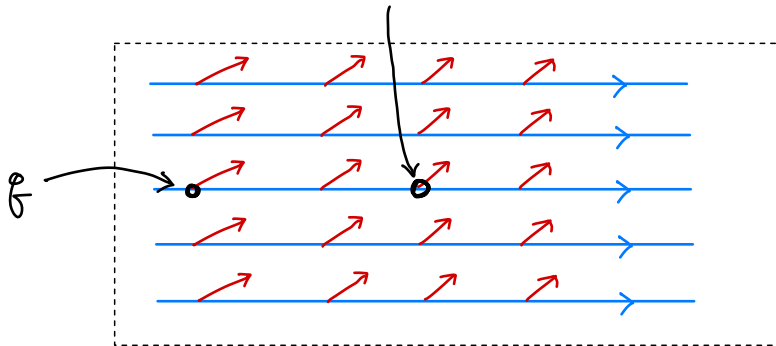
Lie derivative



Lie derivative

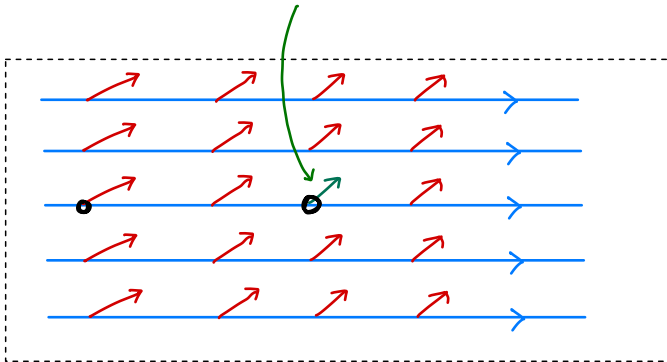
Move $\mathfrak{f}$ by the flow Θ_f .

$$\Theta_f^{(t)}(\mathfrak{f})$$



Lie derivative

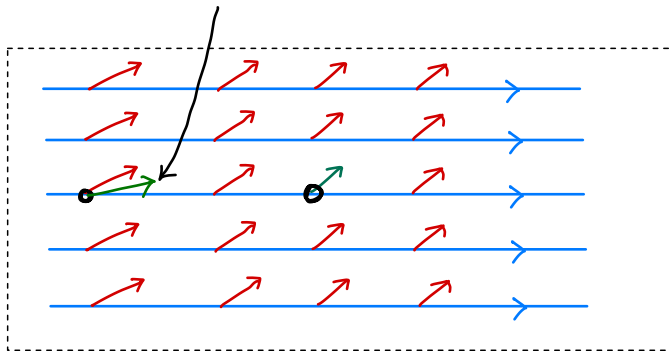
$$g(\theta_p^{(+)}(f))$$



Lie derivative

move $g(\theta_f^{(+)}(p))$ backward.

$$\theta_f^{(-t)}(g(\theta_f^{(+)}(p)))$$



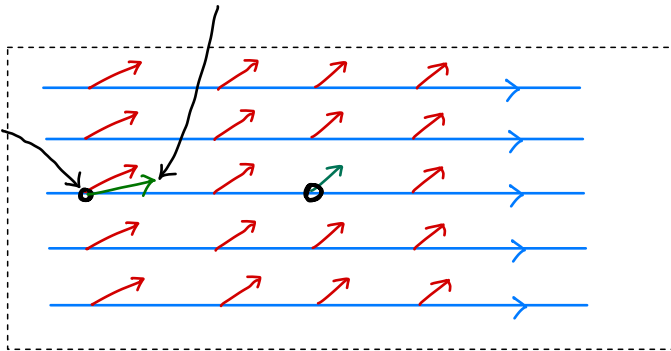
Lie derivative

move $g(\theta_f^{(+)}(f))$ backward.

$$\theta_f^{(-t)}(g(\theta_f^{(+)}(f)))$$

$$\theta_f^{(-t)}(\theta_f^{(+)}(f))$$

$$= f$$

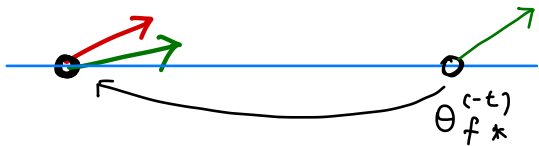


Lie derivative

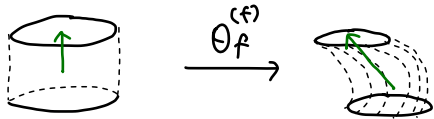


$$(\mathcal{L}_f g)(p) = \lim_{t \rightarrow 0} \frac{\theta_f^{(-t)*}(g(\theta_f^{(+)}(p))) - g(p)}{t}$$

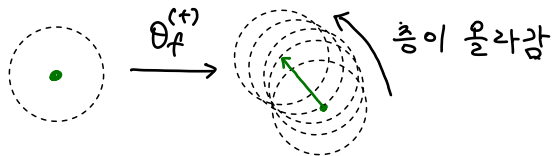
Lie derivative



→ 왜 돌아 가요?



위에서 보면



Lie derivative

Let $F: M \rightarrow M$ be a mapping.

Then $F_{*f}: T_f M \rightarrow T_f M$ is defined by

$$F_{*f}(v) = \left. \frac{d}{dt} \right|_{t=0} F(\alpha(t))$$

where α is a curve in M s.t. $\alpha(0) = f$ and $\alpha'(0) = v$.

Lie derivative

Let $F: M \rightarrow M$ be a mapping.

Then $F_{*f}: T_f M \rightarrow T_f M$ is defined by

$$F_{*f}(v) = \left. \frac{d}{dt} \right|_{t=0} F(\alpha(t))$$

where α is a curve in M s.t. $\alpha(0) = f$ and $\alpha'(0) = v$.

$F_{*f} \simeq \mathbb{R}^3$ 에서 Jacobian 으로 계산할 수 있어요.

Lie derivative

Let us compute $\lim_{t \rightarrow 0} \frac{\theta_f^{(-t)} * (g(\theta_f^{(+)}(p))) - g(p)}{t}$

given $f(p) = (\cos \theta, \sin \theta, 0)$, and

$$g(p) = (\cos \theta, \sin \theta, 1).$$

$$(1) \theta_f^{(+)}(p) = (x + \cos \theta t, y + \sin \theta t, \theta)$$

Lie derivative

Let us compute $\lim_{t \rightarrow 0} \frac{\theta_f^{(-t)} * (g(\theta_f^{(t)}(p))) - g(p)}{t}$

given $f(p) = (\cos \theta, \sin \theta, 0)$, and

$$g(p) = (\cos \theta, \sin \theta, 1).$$

$$(1) \theta_f^{(t)}(p) = (x + \cos \theta t, y + \sin \theta t, \theta)$$

$$(2) g(\theta_f^{(t)}(p)) = (\cos \theta, \sin \theta, 1)$$

Lie derivative

Let us compute $\lim_{t \rightarrow 0} \frac{\theta_f^{(-t)*}(g(\theta_f^{(+t)}(p))) - g(p)}{t}$

given $f(p) = (\cos \theta, \sin \theta, 0)$, and
 $g(p) = (\cos \theta, \sin \theta, 1)$.

(1) $\theta_f^{(+t)}(p) = (x + \cos \theta t, y + \sin \theta t, \theta)$

(2) $g(\theta_f^{(+t)}(p)) = (\cos \theta, \sin \theta, 1)$

(3) $\theta_f^{(-t)*} = \left\{ \frac{\partial \theta_f^{(-t)}}{\partial x_j} \right\}_{\substack{1 \leq i \leq 3 \\ 1 \leq j \leq 3}} = \begin{bmatrix} 1 & 0 & t \sin \theta \\ 0 & 1 & -t \cos \theta \\ 0 & 0 & 1 \end{bmatrix}$

Lie derivative

$$\mathcal{L}_f g = \lim_{t \rightarrow 0} \left(\begin{bmatrix} 1 & 0 & t \sin \theta \\ 0 & 1 & -t \cos \theta \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos \theta \\ \sin \theta \\ 1 \end{bmatrix} - \begin{bmatrix} \cos \theta \\ \sin \theta \\ 1 \end{bmatrix} \right) / t$$

$$= \lim_{t \rightarrow 0} \left(\begin{bmatrix} \cos \theta + t \sin \theta \\ \sin \theta - t \cos \theta \\ 1 \end{bmatrix} - \begin{bmatrix} \cos \theta \\ \sin \theta \\ 1 \end{bmatrix} \right) / t$$

Lie derivative

$$\mathcal{L}_f g = \lim_{t \rightarrow 0} \left(\begin{bmatrix} 1 & 0 & t \sin \theta \\ 0 & 1 & -t \cos \theta \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos \theta \\ \sin \theta \\ 1 \end{bmatrix} - \begin{bmatrix} \cos \theta \\ \sin \theta \\ 1 \end{bmatrix} \right) / t$$

$$= \lim_{t \rightarrow 0} \left(\begin{bmatrix} \cos \theta + t \sin \theta \\ \sin \theta - t \cos \theta \\ 1 \end{bmatrix} - \begin{bmatrix} \cos \theta \\ \sin \theta \\ 1 \end{bmatrix} \right) / t$$

$$= \lim_{t \rightarrow 0} \begin{bmatrix} t \sin \theta \\ -t \cos \theta \\ 0 \end{bmatrix} / t = \begin{bmatrix} \sin \theta \\ -\cos \theta \\ 0 \end{bmatrix}$$

Lie derivative

$$\mathcal{L}_f g(f) = \begin{bmatrix} \sin \theta \\ -\cos \theta \\ 0 \end{bmatrix}, \quad f(f) = \begin{bmatrix} \cos \theta \\ \sin \theta \\ 0 \end{bmatrix}, \quad g(f) = \begin{bmatrix} \cos \theta \\ \sin \theta \\ 1 \end{bmatrix}$$

Lie derivative

$$\mathcal{L}_f g(f) = \begin{bmatrix} \sin \theta \\ -\cos \theta \\ 0 \end{bmatrix}, \quad f(f) = \begin{bmatrix} \cos \theta \\ \sin \theta \\ 0 \end{bmatrix}, \quad g(f) = \begin{bmatrix} \cos \theta \\ \sin \theta \\ 1 \end{bmatrix}$$

Wow! $\text{Span}\{ \mathcal{L}_f g, f, g \} = \mathbb{R}^3$.

Fact

$$\mathcal{L}_f g = [f, g]$$

Rashevskii - Chow theorem

Definition

$\mathcal{F} \subseteq \text{Vect}(M)$ is called **completely nonholonomic**

if $\text{Lie}_{\mathcal{F}}(f) = T_f M \quad \forall f \in M$,

where $\text{Lie}_{\mathcal{F}}(f) = \text{span} \{ [f_N, [\dots, [f_2, f_1] \dots]](f) \mid f_i \in \mathcal{F}, N \in \mathbb{N} \}$.

Rashevskii - Chow theorem

Definition

$\mathcal{F} \subseteq \text{Vect}(M)$ is called **completely nonholonomic**

if $\text{Lie}_{\mathcal{F}}(f) = T_f M \quad \forall f \in \mathcal{F}$.

where $\text{Lie}_{\mathcal{F}}(f) = \text{span}\{[f_N, [\dots, [f_2, f_1] \dots]](f) \mid f_i \in \mathcal{F}, N \in \mathbb{N}\}$.

Example

Our system $\mathcal{F} \subseteq \text{Vect}(\mathbb{R}^2 \times S^1)$ is completely nonholonomic

since $\text{Lie}_{\mathcal{F}}(f) = \text{span}\{[f, g], f, g\} = \mathbb{R}^3 = T(\mathbb{R}^2 \times S^1)$.

Rashevskii - Chow theorem

Theorem (Rashevskii - Chow)

If $\mathcal{F} \subseteq \text{Vect}(M)$ is completely nonholonomic
and M is connected,

then $O_f = M$ for any $f \in \mathcal{F}$ (O_f : orbit)

Rashevskii - Chow theorem

Theorem (Rashevskii - Chow)

If $\mathcal{F} \subseteq \text{Vect}(M)$ is completely nonholonomic
and M is connected,

then $O_f = M$ for any $f \in \mathcal{F}$ (O_f : orbit)

Corollary

자동차는 뱅뱅글글 누빌 수 있다.